08/03/2026

Transfer learning

Pre-training / Fine-tuning / & Parameter Efficient fine Tuning (LoRA, QLoRA)

Transfer Learning

LLM → Pretraining / Supervised fine-tuning / Preference tuning

1. Language Modeling :

Creating LLM → pretrain it / predict the next token

Language modeling

→ base model / pre-trained mod / foundation models.

2. Supervised Fine-Tuning (SFT) Instruction tuned

what is your name and what is your address ?

→ Adapt the base model for your own set of instructions.

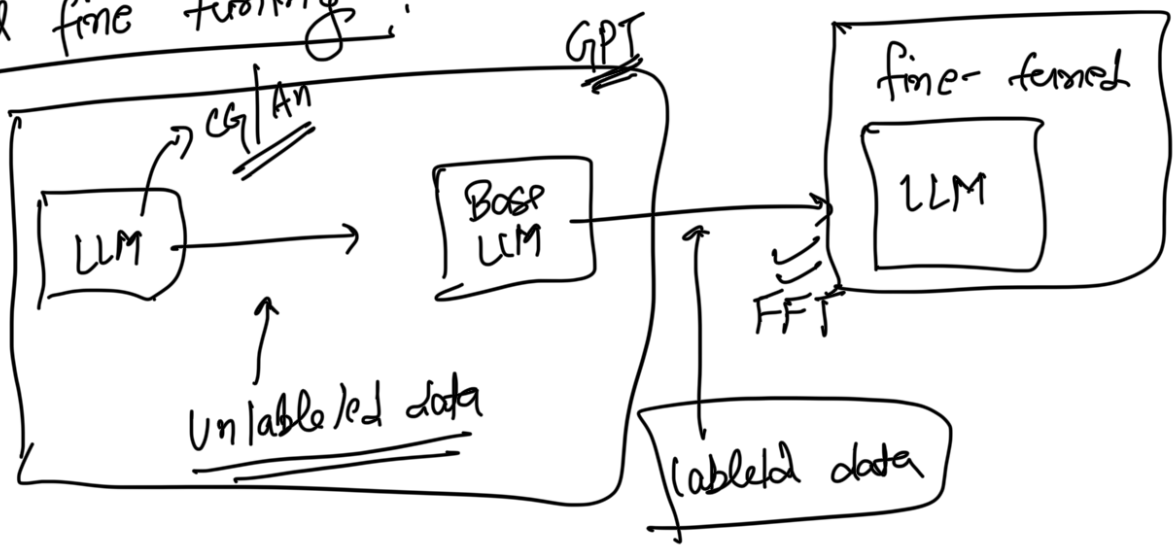
3. Preference Tuning : Improving the quality of the model.] → Even more personalized according to your task.

SFT :

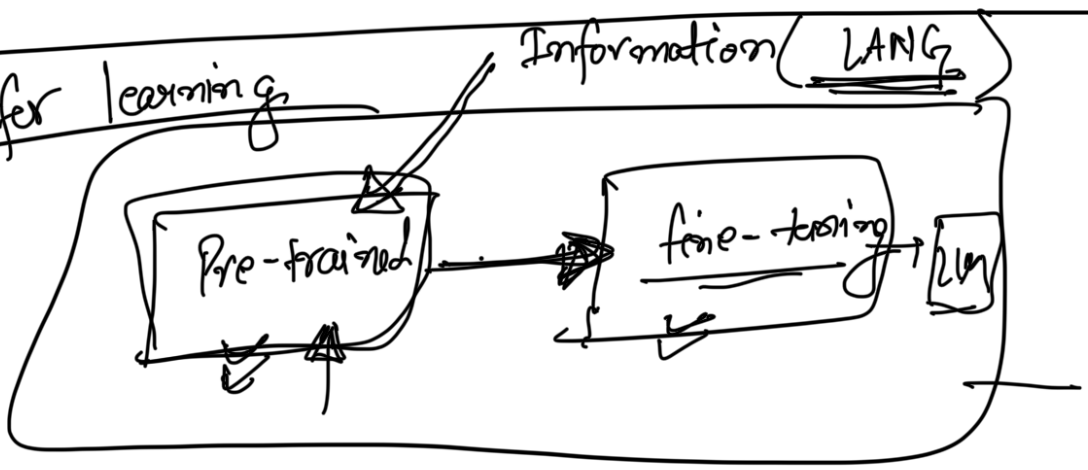
what is 2+3 → [LLM] → 5

what is 2+3 ? → [LLM] → what is 2+3+4 ?

Full fine tuning :

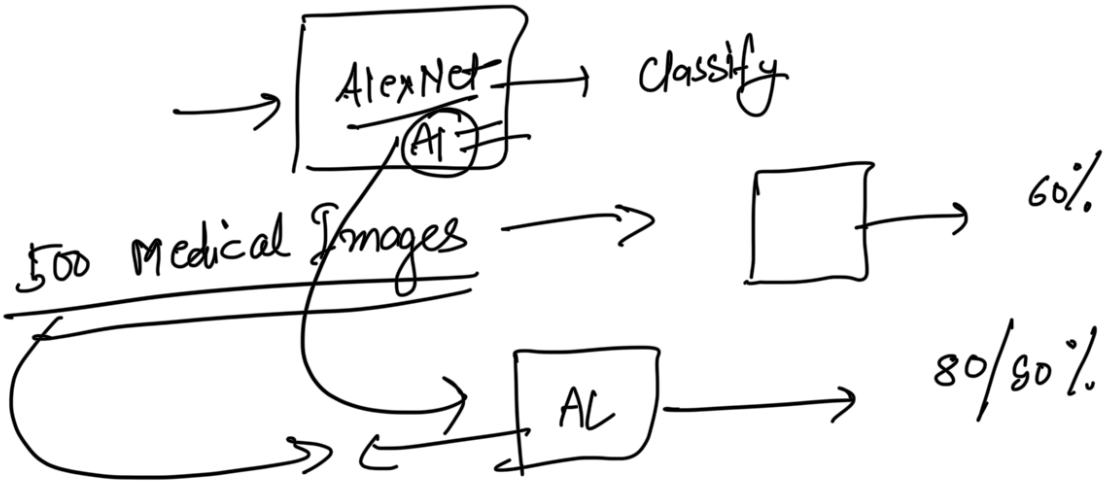


Transfer learning



ImageNet Dataset

1.2 M train images



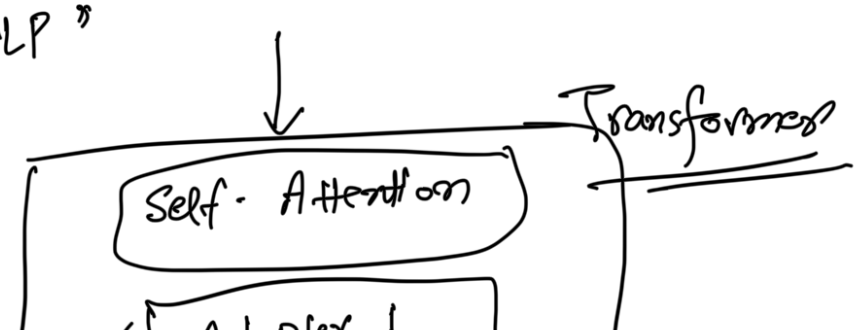
Parameter Efficient Fine Tuning:

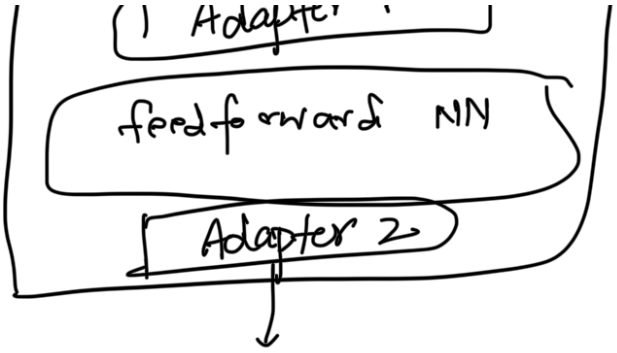
- ↳ cost of training
- ↳ slow training times
- ↳ very high storage

Adaptors (A) (Core - Component)

↳ set of modular components inside the transformer which are tuned to improve the model performance on any specific task.

"Parameter - efficient transfer learning for NLP"




Adapter
feedforward NN
Adapter 2